

Co-phasing of genetic and epigenetic variation with learned error detection improves long-read haplotypes

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Abstract

Long reads allow haplotypes to be phased from a single sample, but read-based phasers weigh every read observation alike, mostly ignore evidence other than single-nucleotide variants and report phase as a binary state, so switch errors pass unnoticed downstream. Here we present LongPhase 2, which co-phases SNVs, small indels, structural variants and allele-specific methylation in one graph, weights each observation by base quality and sequence context, and assigns every phased variant an uncertainty score. A graph neural network then unphases, rather than flips, the variants it predicts to be misphased. On HG002 nanopore data at 10–60× coverage, scored against a telomere-to-telomere truth set, LongPhase 2 makes 2.6- to 5-fold fewer switch errors than WhatsHap and HapCUT2 (1.1- to 1.3-fold under GIAB v4.2.1) and runs six- to eightfold faster at 60×; co-phasing lengthens haplotype blocks by 40%. LongPhase 2 is a drop-in replacement for the LongPhase versions embedded in Clair3, ClairS, Severus and EPI2ME.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Whether two variants act in *cis* or in *trans* decides compound heterozygosity in recessive disease¹, and allele-specific methylation on a phased haplotype reveals its parent of origin². Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals, so a single library yields haplotype blocks orders of magnitude longer than short reads allow^{3,4}. One sample suffices, without the trios, chromosome-conformation or single-cell strand data that chromosome-scale phasing otherwise requires⁵, and without the reference panels on which statistical phasing depends⁶.

Read-based phasing is classically formulated as minimum error correction and solved by dynamic programming or graph partitioning, as in WhatsHap⁷, HapCUT2⁸ and Margin⁹, all of which were designed for single-nucleotide variants (SNVs). We previously introduced LongPhase¹⁰, which recasts phasing as a weighted allele graph, resolves haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{11,12}. Much of the long-read analysis

stack now rests on that graph. Clair3 can use LongPhase, and the somatic callers ClairS and ClairS-TO use it by default, for the phasing that precedes variant calling^{13–16}; the EPI2ME wf-human-variation workflow phases every nanopore genome with it¹⁷; the SV caller Severus consumes its haplotagged alignments¹⁸; and published work on cancer evolution, rapid rare-disease diagnostics and imprinting inherits its haplotype blocks^{19–23} (Supplementary Table 6). An error made by the phaser is therefore inherited, without any warning, by every analysis downstream.

Deployment at this scale has exposed three limitations of current phasers. The first concerns evidence beyond SNVs. Small indels, SVs and 5-methylcytosine (5mC) each carry haplotype linkage that SNVs cannot supply, and that linkage matters most where heterozygosity is low and blocks terminate. Methylation-aware methods such as NanoMethPhase²⁴ and MethPhaser²⁵ nevertheless operate as post-processing on an SNV phaser’s output, published joint phasing of indels and SVs has been restricted to high-accuracy HiFi reads²⁶, and no published method places all four evidence classes in one model. Nor is the extra evidence free: nanopore indel calls are markedly less accurate than SNV calls¹³, so unweighted indel links can introduce errors as well as contiguity. The second limitation, which compounds the first, is that read evidence is weighted uniformly even though the errors that break haplotypes are not uniformly distributed. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions mislead the graph^{27,28}. What follows is the switch error, a flip of haplotype assignment within a block that inverts every inference downstream, from compound-heterozygosity calls to haplotype-aware somatic detection^{15,18}.

The third limitation is that phase is reported as a binary state. Statistical phasers have long attached a posterior probability to every phased genotype⁶, and HapCUT2 prunes variants whose posterior falls below a threshold⁸. That posterior, however, comes from the same likelihood that produced the phase, a model that treats reads as independent and correctly placed; it is not designed to detect the correlated errors that mis-mapped reads and collapsed paralogues introduce. WhatsHap, Margin and LongPhase report no confidence at all, leaving a phase supported by many concordant reads indistinguishable from one supported by two conflicting reads. Learning-based phasers^{29,30} replace the haplotype-assembly algorithm rather than diagnose where phasing goes wrong, and handle only one variant class.

These limitations have been easy to underestimate, because phasers are benchmarked against GRCh38-based Genome in a Bottle (GIAB) truth sets³¹ that exclude most segmental duplications and satellite sequence^{27,28}, the regions in which uniform evidence weighting is expected to fail. The complete diploid assembly of HG002³² now provides a truth set that assesses those regions³³, much as the complete CHM13 reference did for variant calling³⁴.

Here we present LongPhase 2, a germline phaser that co-phases SNVs, small indels, SVs and 5mC in one weighted phasing graph and weights each read observation by base quality and sequence context. Every phased variant receives an entropy-based uncertainty score, and a graph neural network examines the phasing graph around the most uncertain variants; variants it predicts to be misphased are unphased instead of flipped, so genotypes are preserved (Fig. 1). On HG002 nanopore data at 10–60×, scored against the T2T-HG002-derived bench-

mark, LongPhase 2 with neural-network correction makes 2.6- to 5-fold fewer SNV switch errors than WhatsHap and HapCUT2, a margin that narrows to 1.1- to 1.3-fold under GIAB v4.2.1, and runs six- to eightfold faster at 60×. Co-phasing all four variant classes lengthens haplotype blocks by 40% at 60× while making fewer switch errors than uncorrected SNV-only phasing. LongPhase 2 is a drop-in replacement for the LongPhase versions deployed in Clair3, ClairS, Severus and the EPI2ME workflows, and the neural-network correction is an optional sub-command, so those pipelines gain the new phasing without any change to their interfaces.

Results

LongPhase 2 co-phases SNVs, indels, SVs and methylation in one graph and scores every phased variant

LongPhase 2 resolves haplotypes in three stages (Fig. 1). All heterozygous evidence carried by the same long reads, comprising SNVs, small indels, SVs and allele-specific 5mC sites called by the built-in `modcall` module, enters a single weighted phasing graph per chromosome. Within that graph each observation is weighted by base quality and by the reliability of its sequence context, so that homopolymers, tandem repeats and copy-number-altered regions contribute less (Fig. 1a,b). Each phased variant also receives a phasing-uncertainty score, and a graph neural network inspects a local window around every variant whose uncertainty is high (Fig. 1c). Variants predicted to be misphased are unphased instead of flipped, and any haplotype block that becomes disconnected is split, so genotypes are preserved and no new phase assignment is ever introduced. The outputs are phased VCFs for every variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 whole-genome data down-sampled from 10× to 60× coverage (Methods). Throughout, values quoted at 10–20× are means over ten independent down-sampling replicates (nine at 10×), and values quoted at 30–60× come from a single replicate. Phasing accuracy was scored against two independent truth sets: the GRCh38-based GIAB v4.2.1 benchmark, and the benchmark derived from the complete diploid telomere-to-telomere (T2T) assembly of the same individual, T2T-HG002 (referred to here as v5.0q), which extends evaluation into the segmental duplications, satellite arrays and other difficult regions that v4.2.1 excludes. Unless stated otherwise, results refer to the v5.0q benchmark and to SNVs, the only variant class that all compared tools phase.

LongPhase 2 makes 2.6- to 5-fold fewer SNV switch errors than WhatsHap and HapCUT2 and runs six- to eightfold faster

On identical Clair3 variant calls and SNV-only phasing, LongPhase 2 made several-fold fewer switch errors than WhatsHap 2.8 and HapCUT2 1.3.4 at every coverage (Fig. 2). Without the neural network, the switch error rate of LongPhase 2 was 0.046% at 30× and 0.034% at 60×, against approximately 0.115% and 0.085% for WhatsHap and 0.15% and 0.11% for HapCUT2, a 2.5-fold reduction relative to WhatsHap and a 3.2-fold reduction relative to HapCUT2 (Fig. 2b)[the WhatsHap and HapCUT2 rates in this paragraph (approximately 0.115/0.085% at 30/60× and 0.225/0.27% at 10×) are read from the plots; replace them with the `compare-table`

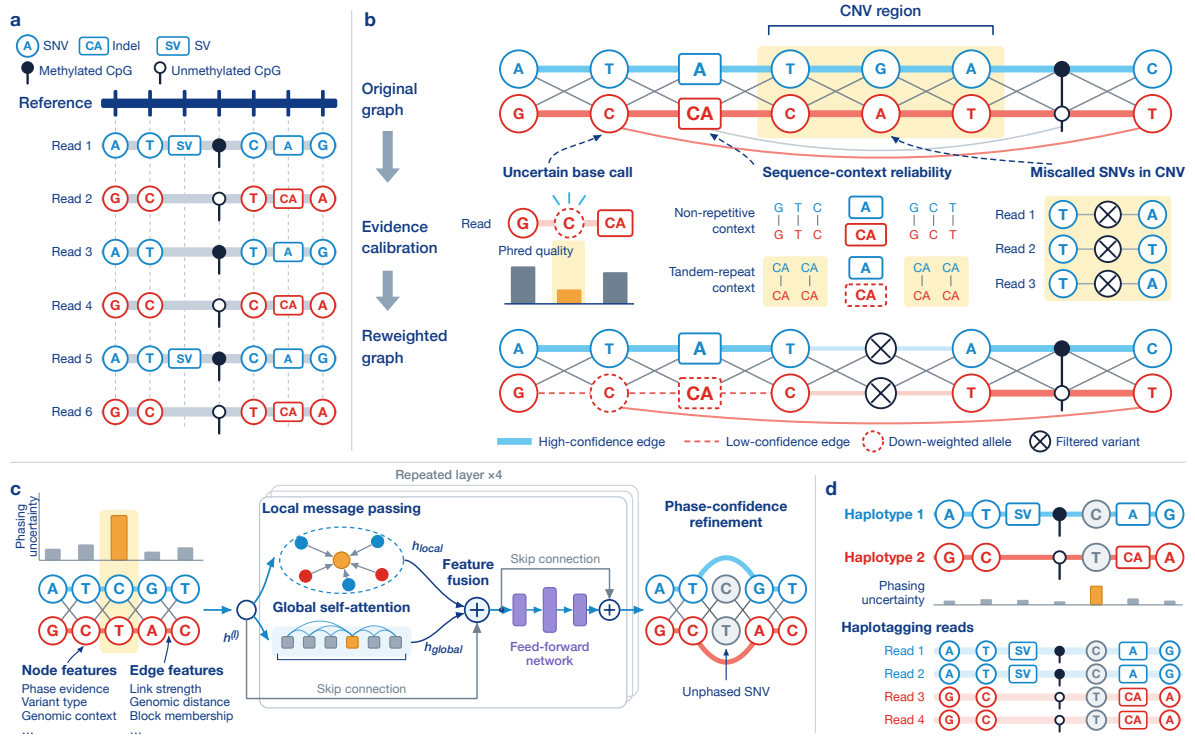


Figure 1: Overview of LongPhase 2. **a**, Heterogeneous molecular evidence (SNVs, small indels, structural variants and 5mC calls) from the same long reads is integrated into a single haplotype-resolution problem. Alleles are coloured by the haplotype of the read that carries them; at the fifth variant two reads (3 and 4) carry the allele of the other haplotype. **b**, Weighted phasing graph. Allele nodes are connected by read-supported edges; evidence is weighted by base quality and sequence-context reliability (homopolymers, tandem repeats, miscalled heterozygous sites in copy-number-altered regions) before multi-neighbour haplotype voting and read-based correction. In the sequence-context inset each context shows the haplotype 1 (blue) over the haplotype 2 (red) sequence around an A/CA indel: the insertion is reliable in unique sequence and down-weighted inside a CA repeat, where its placement is ambiguous. In the reweighted graph, down-weighted alleles are outlined with dashes, their edges become low-confidence (dashed) and miscalled variants in the CNV are filtered. **c**, Uncertainty correction. Each variant receives a phasing-uncertainty score (bars; Methods), and the most uncertain variants (orange) are passed to a graph neural network. Node features (for example phase evidence, variant type and genomic context) and edge features (for example link strength, genomic distance and block membership) of the window around the trigger variant (highlighted) enter two parallel branches: local message passing, in which each node attends to its neighbours in the phasing graph (GATv2), and global self-attention, in which every variant in the window attends to every other (Transformer). The two representations (h_{local} , h_{global}) are fused additively together with a skip connection from the layer input $h^{(l)}$, and passed through a feed-forward network with a second skip connection; the layer is repeated four times and a classifier then labels each variant as correctly phased or misphased. Misphased variants are unphased (grey), preserving the genotype. Here reads spanning the unphased variant still link its neighbours (arcs), so the haplotype block stays intact; a block left disconnected by unphasing is split in two. Layer normalization and the classifier head are omitted from the schematic (Methods; Supplementary Fig. 6). **d**, Outputs: two phased haplotypes carrying the variants of **a** (SNVs, an SV, a small indel and methylated (filled) or unmethylated (open) CpG sites), the variant unphased in **c** (grey) with the per-variant uncertainty score, and reads haplotagged by the alleles they carry.

values, quote the 10× rates, and recompute every fold-change in this paragraph (with the plot-read values and the exact LongPhase 2 rates the 30× and 60× fold-changes both round to 2.5 and 3.2–3.3, so no range is needed)]. Correction reduced the rate further, to 0.031% at 30× and 0.023% at 60×, so that LongPhase 2 with correction made 3.6–3.9-fold fewer switch errors than WhatsHap and 4.7–5.0-fold fewer than HapCUT2 across 30–60×, and 2.6-fold and 3.1-fold fewer at 10×. In absolute terms correction helped most where evidence is sparse: at 10× the rate fell by 0.039 percentage points, from 0.121% to 0.082% (replicate s.d. ≤ 0.002 percentage points), against 0.015 percentage points at 30× and 0.011 at 60×; in relative terms the reduction was close to one-third at every coverage (31–34%). The block-wise Hamming distance, which counts misassigned variants rather than switch points and therefore penalizes long-range flips, showed the same ordering: at 30× the distance was approximately 2.6% for WhatsHap and HapCUT2, 2.4% for LongPhase 2 and 1.8% after correction, and at 10× the distance fell from 7.7–7.8% for the two competing tools to 5.3% (s.d. 0.2) after correction (Fig. 2c).

This gain in accuracy did not cost completeness or haplotype length. All tools phased a similar fraction of benchmark heterozygous SNVs, rising from roughly 78% at 10× to a plateau above 91% from 20× onwards, with WhatsHap phasing about one percentage point more than LongPhase 2 at every coverage (approximately 92.5% versus 91.4–91.6% at 30–60×), consistent with the more conservative read-based correction of LongPhase 2 (Fig. 2a). Block N50 grew almost linearly with coverage for all tools and was highest for LongPhase 2 (approximately 2.9 Mb at 60×, versus 2.75 Mb for WhatsHap and HapCUT2); the neural network shortened blocks by 5–9% at 10–20× (0.84 versus 0.92 Mb at 10×, replicate s.d. 0.01 Mb) and by less than 3% from 30× onwards (2.86 versus 2.91 Mb at 60×) (Fig. 2d). LongPhase 2 was also substantially faster: whole-genome phasing took 3 min at 10× and 15 min at 60×, compared with 26 and 95 min for HapCUT2 and 26 and 115 min for WhatsHap, a 6–8-fold speed-up at 60× (5–8-fold across 30–60×); the neural network added 1–2 min (Fig. 2e)[threads and hardware; exact times, including the 30× times (approximately 12, 65 and 90 min); add peak memory alongside runtime in a Supplementary table, which Methods states was recorded. The repository README tabulates phase of v2.0 at 24 threads as 39 s at 10× and 180 s at 60× with 26.9–52.8 GB peak memory, about five times faster than the 3 and 15 min quoted here from the slides; establish which build, machine, thread count and pipeline stages the figure reports].

A telomere-to-telomere benchmark exposes switch errors hidden by the GRCh38 truth set

Scoring the same phased VCFs against the two truth sets changed the apparent separation between methods by more than fourfold (Fig. 3). Under GIAB v4.2.1, all methods converged to a narrow band of 1,600–2,200 switch errors genome-wide from 30× upwards, with corrected LongPhase 2 lowest (approximately 1,650), followed by uncorrected LongPhase 2 (1,700), WhatsHap (1,800), LongPhase 1.0 (2,000) and HapCUT2 (2,100) (Fig. 3b)[exact counts]. Under v5.0q the same VCFs separated four to five times more widely: at 60× HapCUT2 made approximately 2,330 switch errors, WhatsHap 1,790, LongPhase 1.0 1,380, LongPhase 2 743 and LongPhase 2 with correction 496, and at 10× the corresponding counts were 5,030, 4,390, 4,030, 2,258 and 1,517 (Fig. 3c).

Metrics other than the error counts separate the five methods far less. The fraction of benchmark heterozygous SNVs phased (Fig. 3a) and the phase-block N50 (Fig. 3d), which is benchmark independent, differ little between methods. Hamming distance was also nearly identical under the two benchmarks for HapCUT2, WhatsHap and LongPhase 2 with and without correction (Fig. 3e,f), indicating that the additional v5.0q switch errors are mostly isolated switches rather than long-range flips. LongPhase 1.0 was the exception. Despite an intermediate number of switch errors, LongPhase 1.0 showed a Hamming distance of 6–12% under both benchmarks, which points to long-range flips within blocks. The read-based correction and copy-number-aware filtering introduced in LongPhase 2 remove these.

Co-phasing indels, SVs and methylation lengthens blocks by up to 75% but raises the SNV switch error rate

Joint phasing of all four variant classes extended haplotypes well beyond the SNV-only limit, at a measurable cost in SNV accuracy that correction then more than recovered (Figs. 4 and 5). Indels and methylation sites provide new links across SNV-poor regions, so block N50 at 60× increased from approximately 2.9 Mb for SNV-only phasing to 4.3 Mb with indels, to 3.1 Mb with methylation alone and to 5.1 Mb with all four classes (Fig. 5, right column). At the same time, indel observations are noisier than SNV observations: the SNV switch error rate of uncorrected co-phasing rose from 0.034% (SNV only) to 0.043% (all classes) at 60× and from 0.121% to 0.154% at 10×, while the Hamming distance rose from 1.8% to 7.2% at 60× (Fig. 4e,f). Decomposing the combinations showed that indels carry almost all of the additional error. SNV+SV and SNV+methylation had switch error rates and Hamming distances indistinguishable from SNV-only phasing, while SNV+indel matched the four-class configuration (Fig. 5).

Adding classes did not make the phasing of any single class less complete. The fraction of phased SNVs was unchanged relative to SNV-only phasing (Fig. 4a). The fraction of benchmark heterozygous indels that were phased (of all v5.0q heterozygous indels, including those Clair3 did not call; Methods) rose from 31% at 10× to 45% at 60× and was unaffected by the neural network (Fig. 4b). Of the allele-specific 5mC sites identified by `modcall`, more than 99% were phased at every coverage, and the neural network unphased 1.5–5% of those sites, mostly at low coverage (Fig. 4c). Roughly 68–70% of heterozygous SVs called by Sniffles2 were phased, decreasing slightly with coverage as more low-support SVs were called, and correction removed a further 2 percentage points (Fig. 4d).

Neural-network correction reduces switch errors in every variant-class configuration

The neural network reduced the SNV switch error rate in all five configurations and at every coverage (Fig. 5, left column). For SNV-only phasing the rate fell by about one-third at every coverage (31–34%; for example from 0.034% to 0.023% at 60×); for all four classes it fell by 37–40%, from 0.043% to 0.027% at 60× and from 0.154% to 0.093% at 10× (replicate s.d. 0.002 and 0.001 percentage points), and SNV+indel behaved alike. Corrected co-phasing was therefore more accurate than uncorrected SNV-only phasing at every coverage (Fig. 4e) while retaining substantially longer blocks (Fig. 5, right column). The correction was even more

effective on the Hamming distance of indel-containing configurations, which fell from 7.2% to 3.3% at 60× and from 11.0% to 7.0% at 10× (replicate s.d. 0.4 percentage points) (Fig. 5, middle column), which suggests that a large share of the indel-induced errors are long-range flips the network recognizes from the surrounding vote structure. The price was a moderate reduction in block N50 for indel-containing configurations, from 4.3 Mb to 3.7 Mb (SNV+indel) and from 5.1 Mb to 4.1 Mb (all classes) at 60×, because unphased variants that bridged two blocks split those blocks. For configurations without indels the N50 was almost unchanged. On balance, co-phasing remains worthwhile: even after correction, four-class co-phasing yielded blocks approximately 40% longer than SNV-only phasing at 60×, together with a lower switch error rate than uncorrected SNV-only phasing.

Unphased variants are predominantly unsupported heterozygous calls

To see what the network actually removes, we classified every SNV that LongPhase 2 phased and the network subsequently unphased (SNV-only phasing, single replicate, seed 1) against the v5.0q benchmark (Fig. 6). The network unphased approximately 60,000 SNVs at every coverage, but 86–96% of those SNVs were calls that the benchmark does not support as heterozygous, either because the site is absent from the benchmark (80–94%) or because the benchmark calls the site homozygous. Only 4.0% (2,411 variants) at 60× and 14% at 10× were genuine heterozygous sites (Fig. 6a). Little real phasing information is therefore lost.

Cross-tabulating each unphased variant by read depth and by phasing informativeness, the number of informative reads linking each variant to its neighbours, showed that genuine heterozygotes were removed overwhelmingly at normal depth with weak phasing evidence (61–81%), whereas benchmark-absent calls shifted towards low depth as coverage increased (12.5% at 10× to 58.6% at 60×), as expected in hard-to-map regions where reads are lost to alternative loci (Fig. 6b). High depth alone never caused a variant to be unphased. Among the genuine heterozygotes that were removed, 43% at 10× had a phasing entropy of exactly 0, that is, unanimous but sparse haplotype votes or, at the first variant of a block, no votes at all (Methods), compared with 17% of benchmark-absent calls, and this share fell to 22% at 60× as evidence accumulated; a further 21–25% of genuine heterozygotes and 16–18% of benchmark-absent calls had near-equivocal votes at every coverage (Fig. 6c) [PE ≥ 0.8 prevalences read from Fig. 6c; replace with exact values]. The vote balance at the variant itself would therefore have missed most of the removed heterozygotes; the network, which sees the votes of all neighbours in the window, did not. Clustering within 50 bp and low genotype quality, by contrast, were present in 46–49% of benchmark-absent calls at every coverage and were far less frequent among genuine heterozygotes, which marks those calls as variant-caller artefacts rather than phasing failures.

Correction therefore loses little true phasing information, about 2,400 SNVs at 60× out of 2.19 million phased benchmark SNVs [the compare tables put the loss of phased benchmark SNVs at 60× at 2,055 (2,193,366 before, 2,191,311 after correction), against the 2,411 genuine heterozygotes counted in Fig. 6a. compare counts a benchmark SNV as phased only when a query record matches its position, REF and first ALT exactly and both genotypes are 0/1 or 1/0 (CompareProcess.cpp:360–364, 455–465), so the 356 surplus must come from a looser rule in the Fig. 6 classification (position-only matching, or benchmark records that are multi-allelic or

heterozygous between two ALT alleles, which compare ignores); align the Fig. 6 script with that rule or state the difference]. Most removals are false heterozygous calls whose spurious phase would otherwise propagate into haplotagged reads and downstream haplotype-aware callers.

Discussion

LongPhase 2 rests on two ideas. The first is that every class of heterozygous evidence a read carries belongs in one haplotype model, provided that each observation is weighted by how far it can be trusted; the second is that a phaser should report, variant by variant, how far its output can be trusted. The first extends an observation made independently for methylation^{2,24,25} and for SVs^{10,26}, namely that SNVs alone leave haplotype blocks stranded wherever heterozygosity is low. No previously published method combines all four classes, and HiPhase, which phases indels and SVs jointly, is confined to HiFi reads²⁶. Our results also show that on nanopore reads the extra evidence is not free. Co-phasing all four classes, with no post-processing step, raises block N50 above SNV-only phasing by 75% at 60× (5.1 versus 2.9 Mb), but indels also raise the SNV switch error rate, and only with correction is four-class co-phasing both longer (40%) and more accurate than SNV-only phasing. The second idea gives downstream analyses a measure of phase reliability comparable to the genotype quality attached to variant calls, and to the posteriors that statistical phasers have long reported for imputed haplotypes⁶. That measure replaces the binary phased-or-unphased state that most read-based tools, ours included, have reported until now.

Perhaps the most consequential finding here concerns evaluation rather than algorithms. Scored against two truth sets, the same phased VCFs on the same reference give different error magnitudes and, for some tools, a different ordering. GIAB v4.2.1³⁵ excludes most segmental duplications, satellite arrays and large tandem repeats from assessment^{27,28}; the benchmark derived from the complete diploid T2T-HG002 assembly^{32,36}, projected onto GRCh38, adds some 700 Mb of autosomal sequence that v4.2.1 never assessed, most of it satellite and segmentally duplicated³². Every tool accumulates more switch errors under the T2T-based benchmark, in rate as well as in absolute terms, and the spread between tools widens from under 30% to three- to fivefold. That widening is what one would expect if mis-mapped reads, homopolymer errors and collapsed paralogues create systematically wrong linkages rather than random noise, which is what the evidence weighting in LongPhase 2 targets; we have not stratified switch errors by genomic context to test that directly. The ranking can also invert. LongPhase 1.0 made more switch errors than WhatsHap under v4.2.1 and fewer under the T2T-based benchmark. Almost every published comparison of phasers^{7–10,25,26} used GRCh38-based GIAB releases and so describes the easier part of the genome, much as variant-calling benchmarks did before complete references exposed the rest^{27,34}. Two observations suggest that the extra errors are genuine and not artefacts of a draft truth set. Hamming distance is nearly unchanged between benchmarks for the four current tools, which points to isolated switches and not to the long-range disagreements a mis-projected benchmark would create. The advantage of LongPhase 2 over WhatsHap and HapCUT2 also persists under v4.2.1, though with a smaller margin. The T2T-HG002 benchmark is nonetheless still being refined, and residual discordance in the hardest regions may be benchmark rather than phasing error; comparisons should therefore report both

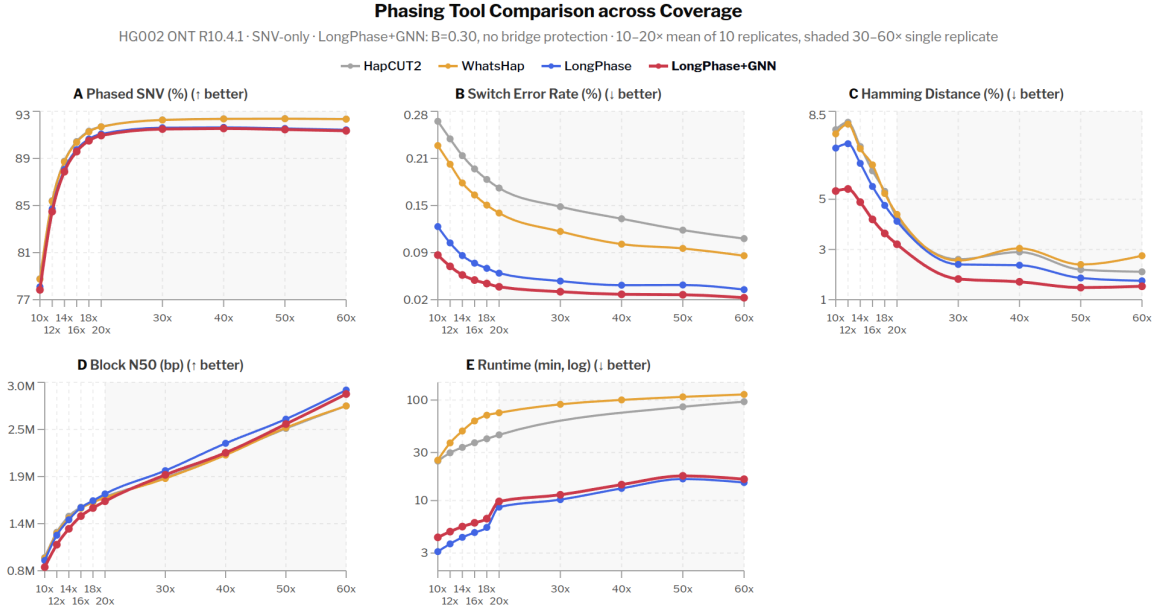


Figure 2: **SNV phasing accuracy, block length and runtime compared with WhatsHap and HapCUT2.** HG002 nanopore R10.4.1 reads down-sampled to 10–60×; SNV-only phasing of identical Clair3 calls; scored against the benchmark derived from the complete diploid T2T-HG002 assembly (v5.0q). Points at 10–20× are means of independent down-sampling replicates ($n = 10$, except $n = 9$ at 10×); the shaded region (30–60×) is a single replicate. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Wall-clock runtime (log scale, $[t]$ threads, whole genome); LongPhase 2 with correction includes the neural-network stage. Grey, HapCUT2 1.3.4; orange, WhatsHap 2.8; blue, LongPhase 2; red, LongPhase 2 with neural-network correction.

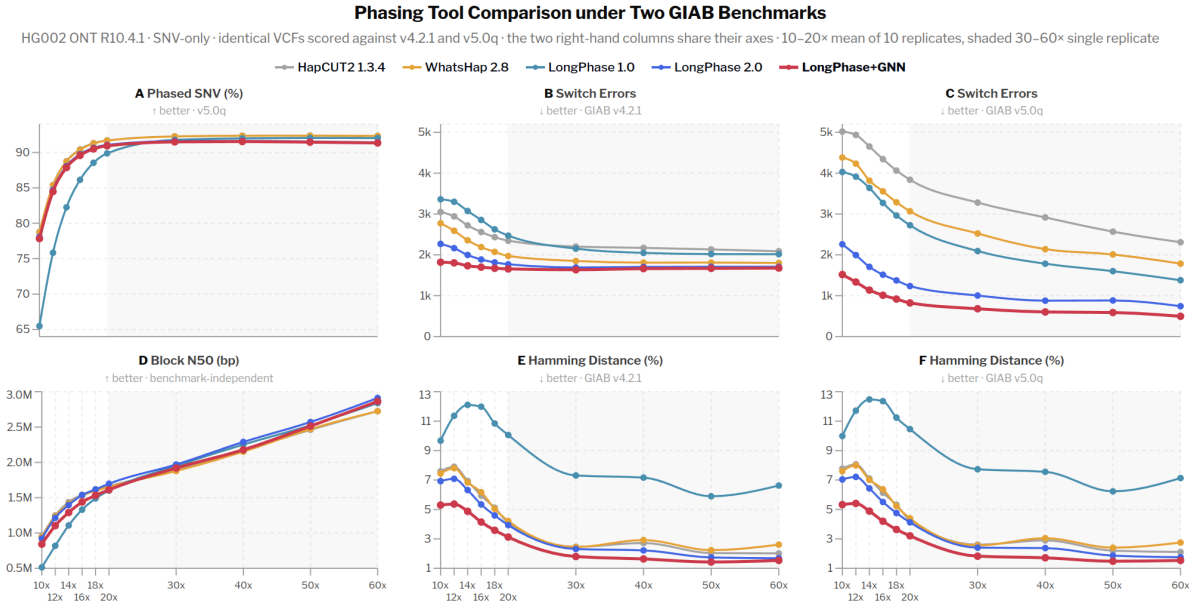


Figure 3: **The same phased VCFs scored against the GRCh38-based GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmarks.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2. Only the truth set differs between the two columns; v5.0q additionally assesses the segmental duplications, satellite arrays and other repeat-rich sequence that v4.2.1 excludes. **a**, Fraction of benchmark heterozygous SNVs phased (v5.0q). **b,c**, Genome-wide switch-error counts under v4.2.1 (**b**) and v5.0q (**c**); the two panels share axes. **d**, Phase-block N50 (benchmark independent). **e,f**, Hamming distance under v4.2.1 (**e**) and v5.0q (**f**). Grey, HapCUT2 1.3.4; orange, WhatsHap 2.8; teal, LongPhase 1.0; blue, LongPhase 2; red, LongPhase 2 with neural-network correction.

truth sets and stratify switch errors by genomic context^{28,37,38}.

The neural network was designed to complement the graph algorithm, not to replace it. Where Ralph²⁹ and GCphase³⁰ learn haplotype assembly itself and must scale to whole chromosomes, LongPhase 2 confines learning to windows around high-entropy variants; inference costs one to two minutes per genome and fewer than 700,000 parameters compile into the binary. The closest precedent is the likelihood-based pruning of HapCUT2⁸, which withholds variants whose posterior under the phasing model falls below a threshold. Our correction conditions on something else. In place of the likelihood that produced the phase, a model that treats reads as independent and correctly placed, the network reads the structure of the evidence across a window: vote totals, entropy, sequence context, graph topology and the concordance of neighbours. That is where we expect correlated errors from mis-mapped reads and paralogues to leave a trace, although we have not compared the two approaches at matched numbers of withheld variants. Because the correction only withholds phase, one might object that fewer switch errors simply reflect a smaller denominator. Three observations speak against that reading. The rate is computed per assessed pair, and of the roughly 60,000 SNVs unphased at 60× only 2,411 are heterozygous in the benchmark and enter that denominator, far too few to explain the fall from 743 to 496 genome-wide switch errors. Block N50 is almost unaffected (2.86 versus 2.91 Mb), which pruning informative links could not achieve. And the phased fraction after correction stays within about a percentage point of WhatsHap (91.3–91.5% versus approximately 92.5% at 30–60×). At 10× the trade is less favourable, with 14% of what is unphased being genuine heterozygotes and N50 falling from 0.92 to 0.84 Mb. Even there the trade remains worth making, because 86–96% of what is removed are calls the benchmark does not support as heterozygous, mostly caller artefacts at clustered or low-quality sites whose spurious phase would otherwise reach haplotagged reads and the somatic callers that consume those reads^{15,16,18}.

Several limitations remain. The network was trained and evaluated on one individual, HG002, sequenced on a single nanopore chemistry. Its labels came from the same benchmark, and its training windows from the same down-sampling replicates, that the evaluation uses; only three chromosomes were held out, and no training window was drawn above 20× (Methods), so an evaluation on an independent sample and benchmark is still outstanding. Two choices bound the cost of a miscalibrated model, although neither is evidence of generalization: the features are generic (vote counts, entropy, sequence context, graph topology; no sequence embeddings, no sample-specific statistics), and the output is restricted to withholding phase, so such a model costs contiguity and cannot invent haplotypes. Transfer to PacBio HiFi, duplex chemistry and non-human genomes is untested; the graph algorithm supports both platforms, but for the network reported here cross-platform performance is an expectation, not a result. Indel co-phasing is a trade-off we have accepted rather than solved. Adding indels lengthens blocks but raises the uncorrected SNV switch error rate (0.034% to 0.043% at 60×), and is worth taking only because correction more than repays the loss, leaving four-class co-phasing both more accurate than uncorrected SNV-only phasing and 40% longer in block N50. Better indel calling^{14,39} would remove the trade-off altogether, and SNV-only phasing remains available in the meantime. The phased fraction of SVs and indels is bounded by the upstream callers, and 5mC co-phasing requires modification-aware base calling, whose accuracy varies between callers

and chemistries⁴⁰, and MM/ML tags retained through alignment. Finally, LongPhase 2 assumes a diploid genome with balanced allele fractions.

Trio assembly^{41,42}, Hi-C⁴³ and Strand-seq⁵ reach chromosome scale but need extra samples or libraries, and imputation^{44,45} depends on panels still sparse for structural variation and many populations; LongPhase 2 needs one sample and one library, and per-variant phase reliability makes the resulting haplotypes a natural input to hybrid read-plus-panel strategies^{6,38}. As a drop-in replacement in pipelines already built on LongPhase^{13,15,17,18}, LongPhase 2 delivers the new annotations without interface changes, and those annotations have concrete uses. A compound-heterozygous diagnosis rests on a single *trans* call between two variants in one gene^{1,46}; the entropy of those variants and of the links between them turns that call into a quantity a diagnostic report can qualify, flagging the cases that warrant a confirmatory assay. Allele-specific methylation analyses currently treat every haplotagged read alike, whereas weighting reads by the confidence of the variants that tagged them would keep imprinting and *cis*-regulatory signals² from being diluted by thinly supported blocks. Haplotype-aware somatic callers^{15,18} likewise accept inherited phase without qualification, and propagating entropy would let them discount the loci at which switch errors concentrate. Realizing any of this will require downstream tools to read the new fields, much as they came to read genotype quality.

Methods

Overview of LongPhase 2

LongPhase 2 takes as input long-read alignments (BAM or CRAM), a reference genome and heterozygous variant calls, and produces phased VCF files for every variant class together with haplotagged alignments (Fig. 1). The pipeline comprises four sub-commands implemented in a single C++ binary linked against HTSlib⁴⁷: **modcall** identifies allele-specific 5-methylcytosine (5mC) sites from the MM/ML modification tags of the alignments; **phase** co-phases SNVs, small indels, SVs and 5mC sites in one weighted phasing graph per chromosome and emits per-variant phasing diagnostics; the optional **gnn** sub-command applies a graph neural network to the diagnostics and removes phase assignments that are predicted to be erroneous; and **haplotag** assigns each read to a haplotype using the phased variants. A fifth sub-command, **compare**, implements the evaluation metrics (below). **phase**, **modcall** and **compare** are parallelized over chromosomes with OpenMP; **gnn** distributes trigger windows over a pool of C++ threads, sharing one immutable copy of the network weights. The graph phasing algorithm extends the design of LongPhase 1¹⁰; the sections below describe the current algorithm, with the implementation-level detail of the individual modules in Supplementary Methods 1–6; command-line parameters and their defaults are documented in the repository (Code availability). LongPhase 2 is a germline, diploid phaser.

Input variant calls

Heterozygous SNVs and small indels were taken from a standard long-read small-variant caller (Clair3¹³ in this study); homozygous and filtered records are carried through to the output unchanged but do not enter the graph. With **-indels**, small indels were phased alongside SNVs;

indels with `QUAL` below a user threshold (`-indelQuality`, default 0, i.e. disabled) were excluded and marked in the output `FILTER` column. Each indel was additionally annotated as lying in a tandem repeat, because indel calls in repetitive sequence are the class most often misplaced by the caller and, once misplaced, link the wrong alleles. The dinucleotide immediately 3' of the indel position was read from the reference, and the indel was flagged when that dinucleotide recurred in five consecutive copies. Only dinucleotide units are considered; extending the test to trinucleotide units was evaluated during development, did not reduce switch errors consistently across coverages and was therefore not adopted. Flagged indels were not discarded but retained as phasing evidence, casting votes at one-tenth weight (see below), so that they can still bridge SNV-poor regions without dominating a contested decision. SVs were taken from Sniffles2¹¹ or cuteSV¹² run with read-name reporting (`-output-rnames` or `-report_readid`); homozygous SVs, SVs whose breakpoint coincides with an SNV, and SVs with duplicated positions were excluded. For each retained SV we recorded the set of supporting read names and the SV length.

Calling allele-specific methylation (`modcall`)

Base-level modification probabilities (Supplementary Fig. 7) were read from the MM/ML tags produced by Dorado⁴⁸ and carried into the alignments by `minimap2 -y`⁴⁹; PacBio HiFi reads carry the same tags but were not used in this study. For each CpG on each read, the base was labelled methylated if the 5mC probability was ≥ 0.8 (`-modThreshold`) and unmethylated if that probability was ≤ 0.2 (`-unModThreshold`); intermediate values were treated as noise. Only the 5mC probability (modification code `m`) was read (`ModCallParsingBam.cpp:169`). 5hmC calls (code `h`) were ignored, so under a 5mCG_5hmCG basecalling model a cytosine called as 5hmC was counted as unmethylated whenever its 5mC probability was ≤ 0.2 . Reads on the reverse strand were mapped to the forward-strand CpG coordinate and the two strands of each CpG were genotyped jointly. Each read was thereby assigned to one of three states at the site, and a site was called heterozygous (allele-specifically methylated) when the two states were balanced and the ambiguous fraction small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6 \text{ (-heterRatio)}, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2 \text{ (-noiseRatio)}, \quad (1)$$

where M_R , U_R and A_R are the numbers of methylated, unmethylated and ambiguous reads; otherwise the site was called homozygous methylated or unmethylated and excluded from phasing. Only CpG dinucleotides were considered; modification calls at non-CpG cytosines were discarded.

A site passing equation (1) is balanced but not necessarily allele-specific, because a site at which methylation varies stochastically between reads satisfies the same inequalities. Candidate sites were therefore required to co-segregate with a neighbouring heterozygous marker. Writing RR , RA , AR and AA for the numbers of reads carrying each of the four combinations of the two states at a pair of sites, a pair was accepted as linked when

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9 \text{ (-connectConfidence)}, \quad (2)$$

that is, when nearly every read supports one of the two possible pairings rather than a mixture. Equation (2) is the criterion the phasing graph itself applies to a pair of alleles, so the two methylation states enter the graph on the same footing as reference and alternate alleles and **phase** requires no methylation-specific machinery.

Linking was applied in two modes: *SNV-anchored*, used whenever an SNV VCF was supplied, in which a candidate CpG must satisfy equation (2) against an SNV among the next 20 variants (**-connectAdjacent**); and *methylation-only*, used where no informative SNV is in range, in which the same criterion is applied between two candidate CpGs. Candidates that never satisfied it were held back as weak points and admitted only on a later pass against an already accepted CpG. Runs of consecutive CpG coordinates were then merged into a single locus represented by the first position, which consolidates the forward- and reverse-strand calls of a run into one read list and removes the need for **phase** to reason about strand. The resulting VCF records the methylated and unmethylated read names (MR, NR), the read strand and the per-site depths, and is passed to **phase** through **-mod-file**. The depth bound on the neighbour scan, the weak-point expansion and the rationale for merging are given in Supplementary Method 1. The values quoted here are the compiled-in defaults.

Extraction of read-level allele observations

For each chromosome, all alignments with mapping quality ≥ 1 (**-mappingQuality**) that were not unmapped, secondary or marked as optical/PCR duplicates were streamed once and, for every variant they overlap, converted into an allele observation with an associated quality. Supplementary alignments were retained as phasing evidence, which is why the overlapping-alignment filter below is required; this differs from the stricter filters applied inside **modcall** and **haplotag**. SNV alleles were read from the aligned base and inherited its Phred base quality; indel alleles were called from the CIGAR string; SV alleles were assigned by scanning a window of 20 CIGAR operations (**-svWindow**) around the SV start for an insertion or deletion whose length is within 10% of the SV length (**-svThreshold**), and reads named by the SV caller were also assigned the ALT allele; and 5mC alleles were taken from the **modcall** read lists. Soft- and hard-clipping operations longer than 5 bp were counted per reference position, separately for clips at the front (left) and back (right) end of an alignment, as evidence of alignment breakpoints for copy-number detection (below). Supplementary Method 2 gives the per-class extraction rules in full.

Three filters were applied before graph construction. (1) *Homopolymer SNV filter*, applied under **-ont** only: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV was removed, because such pairs are dominated by systematic homopolymer-length errors in nanopore reads. (2) *Overlapping-alignment filter*: when two alignments of the same read overlapped by more than 20% of their combined span (**-overlapThreshold**), the shorter alignment was discarded. (3) *Copy-number-aware filter* (Supplementary Fig. 1e,f). Collapsed paralogues and copy-number-altered segments recruit reads from more than one locus, so that paralogue-specific differences are called as heterozygous variants and link alleles that are not on the same chromosome. Such segments are bounded by alignment breakpoints. Two boundary shapes were detected from the per-position front- and

back-clip counts by a three-state scanner: a *step* boundary, characteristic of a copy-number transition, declared on a cluster of at least five front-clips at one position, and a *ramp* boundary, characteristic of the progressively nested breakpoints left by breakage–fusion–bridge cycles, declared where front-clips merely exceed back-clips. Both active states carried a running excess of front- over back-clips, extended the candidate interval by a 30 kb look-ahead while that excess was large, and closed on a back-clip threshold scaled to the strength of the breakpoint that opened the interval; candidates longer than 200 kb, or in which the excess fell to zero, were abandoned. Supplementary Method 2 gives the state machine, its five constants and the closing rules in full.

Within each interval, every read was scored by its *mismatch load* R_m , the number of heterozygous sites inside the interval at which the read carries the alternate allele, and every heterozygous variant v in the interval was then scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where $\bar{R}_m^{\text{ALT}}$ and $\bar{R}_m^{\text{REF}}$ are the mean loads over the reads supporting the alternate and the reference allele of v . Variants with $\text{MR}(v) \geq 0.7$ were removed from all reads; Supplementary Method 2 explains why the statistic separates genuine heterozygous sites from paralogue-specific differences.

Weighted phasing graph

Each heterozygous variant v contributes two nodes (Supplementary Figs. 1–2), v^{ref} and v^{alt} . For every read, the observed alleles were sorted by position and each allele node was connected to the allele nodes of the next $k = 35$ variants on the same read (`-connectAdjacent`); no distance limit is applied to these edges or to the votes cast along them, so a single read can link variants more than 300 kb apart. The `-distance` parameter (300 kb) acts instead on the variant sequence: a variant whose next heterozygous variant lies further away than this is skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy, so that a block cannot be carried across a gap in heterozygosity. Each read contributes a weight of 1 to the edge if both of the two observations have Phred quality ≥ 12 (`-baseQuality`) and a weight of 0.1 (`-edgeWeight`) otherwise. Observations without a base quality were assigned a fixed one (`PhasingGraph.cpp:838–862`): 60 for indel and 5mC observations and for SV observations carrying the ALT allele, and 30 for SV observations on reads that neither the CIGAR scan nor the caller’s read list assigned to the ALT allele, which are taken to carry the reference allele. Both values exceed the default base-quality threshold, so at default settings every indel, SV and 5mC observation contributes a weight of 1; the lower quality of reference-allele SV observations reduces their weight only when `-baseQuality` exceeds 30. For a pair of variants (u, v) this yields four accumulated weights $w_{rr}, w_{ra}, w_{ar}, w_{aa}$, the total support for the two allele pairings $P = w_{rr} + w_{aa}$ (*cis* between reference alleles) and $Q = w_{ra} + w_{ar}$, and their similarity $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. For the current variant u , each of its k downstream neighbours v received a vote for the pairing with the larger support (P versus Q). A pair was declared uninformative and cast no vote if $s > 0.7$ (`-edgeThreshold`); for SNV-5mC pairs the threshold was 0.3, and such pairs cast no vote at all when their total accumulated support was below 1. Votes were weighted: 1 by default; 20 when the pairing was near-unanimous ($s \leq 0.1$ with total support ≥ 1 , or one pairing accumulated less than 1 unit of support while the other accumulated at least 1); and 0.1 when the voting variant was a tandem-repeat indel. [The weight-20 branch in `PhasingGraph.cpp` was chained off `if(debug)` at `cc17fb1`; the fix is applied in the working tree of the source repository (uncommitted as of 2026-09-20) and must be committed before the release. `findBestEdgePair` still takes an unused `isONT` argument.] The vote was translated into a haplotype vote for v using the already assigned haplotype of u , and the two weighted haplotype totals $h_1(v)$ and $h_2(v)$ were accumulated from all upstream voters. A guard against a single erroneous ultra-long read counted, among the votes received by v , those whose total support was at most 1 read-equivalent; when more than three such votes existed and at least one qualifying vote remained, h_1 and h_2 were recomputed from only those votes with $s < 0.2$, weight ≥ 1 and a voting variant that is not a small indel. Variant v was assigned to the haplotype with the larger total. When $h_1 = h_2$ (including the case of no votes at all) and v lay beyond the last position reached by any accepted edge, a new phase block was opened at v ; a tie at a position already covered by an accepted edge left the previous block intact and the tied variant unphased, casting no votes. Blocks were identified by the position of their first variant, which defines the `PS` tag. The six constants of this rule were fixed by a sweep on HG002 at 10–60 $\times$ before the experiments reported here and were not re-tuned afterwards; Supplementary Method 3 states them and Supplementary Fig. 9 shows the sweeps.

Read-based correction

After the initial assignment (Supplementary Fig. 3), every read was assigned to a haplotype by weighted majority over its phased alleles, with SNV and SV alleles contributing 1, indel and tandem-repeat-indel alleles 0.1, and 5mC alleles nothing; a read was tagged only if the majority fraction exceeded 0.65 (`-readConfidence`) and the total weighted evidence exceeded 1. The allele counts of the two read populations were then used to re-derive each variant's phase: the fraction of tagged reads consistent with the assignment $\rho = \max(n_1, n_2)/(n_1 + n_2)$, where n_1 and n_2 count haplotype-1 reads carrying the reference allele plus haplotype-2 reads carrying the alternate allele and vice versa, had to exceed 0.75 (`-snpConfidence`); otherwise the variant was left unphased. This second pass removes assignments driven by a small number of chimeric or mis-mapped reads.

Phasing entropy and graph export

For each phased variant we recorded the two weighted haplotype totals h_1 and h_2 (after the single-read guard, where that guard fired) and their normalized Shannon entropy

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}, \quad (4)$$

which ranges from 0 (unanimous votes) to 1 bit (equal support for both haplotypes), with the convention $0 \log_2 0 = 0$. PE is set to 0 when $h_1 + h_2 = 0$, so a value of exactly 0 denotes either unanimous support or no votes at all; the two cases are distinguished by H1+H2, and the second arises only at the first variant of a phase block, the one position at which a variant without votes is phased (Supplementary Fig. 2f). PE, h_1 and h_2 are written to the output VCF as the INFO fields PE, H1 and H2. With `-dot`, the per-chromosome phasing graph is written in DOT format as `<prefix>.<chrom>.dot`, one directed edge $u^a \rightarrow v^b$ per accepted voting pair with the haplotype weight accumulated at u at the time the vote was cast as label; these files are the input of the GNN module.

Graph neural network for phase-error detection

The graph neural network (GNN) operates on local windows of the phasing graph and predicts, for each variant, the probability that its phase assignment is wrong (Supplementary Figs. 4–6). Only records that are heterozygous and phased in the input VCFs enter the module. Variant records from the SNV/indel, SV and 5mC VCFs were merged and sorted by position so that windows span all co-phased variant classes, and a window was opened around every such variant with $\text{PE} \geq 0.8$ (`-pe-threshold`), extending 20 variants (`-window`) on each side of the trigger in that merged list. A window therefore contains at most $2 \times 20 + 1 = 41$ variants and 82 allele nodes. Windows with more than 256 nodes are skipped; at the default window size this cap is never reached, and binds only for `-window` greater than 63.

Node and edge features. Each allele node carries 31 features in four groups: phasing evidence (PE, the genotype and allele indices, the trigger and phase-set indicators, the signed offset from the trigger normalized by the largest absolute offset in the window, the two weighted haplotype totals h_1 and h_2 , and six transforms of them); genomic context, computed from the reference when `-r` is given and zero otherwise (homopolymer length, GC content, tandem-repeat unit count, local sequence entropy, variant density and distance to the nearest indel); graph structure (out-degree, phase-set size, log offset from the trigger, dispersion of the outgoing edge weights and a bridge-vertex indicator); and four one-hot variant-class indicators. Each edge carries six features derived from the DOT vote weight w , the genomic distance, the allele-index match and the phase-set relation. Edges are restricted to pairs with both endpoints inside the window, symmetrized, and augmented with one self-loop per node whose features are the mean of its non-self incident edges. Supplementary Tables 1 and 2 give every feature with its exact definition, normalization and cap, and Supplementary Method 4 gives the transforms and the provenance of h_1 and h_2 .

Architecture. The network follows the general, powerful, scalable (GPS) graph-transformer

recipe⁵⁰. Node features were standardized by a batch-normalization layer whose statistics are frozen at inference, projected linearly to a 128-dimensional embedding and passed through a GELU non-linearity. Four identical layers follow, each combining a local branch, a four-head GATv2 message-passing layer over the window’s edges^{51,52}, with a global branch, four-head scaled dot-product self-attention over all nodes of the window⁵³. The two branch outputs are added to the residual stream and layer-normalized, then passed through a two-layer feed-forward block (128→256→128, GELU) with a second residual connection and layer normalization. The final node embedding is concatenated with two edge-context scalars and passed through a two-layer classifier (130→128→2) with softmax output.

The model comprises 687,358 float32 values (2,749,432 bytes). Inference is implemented in plain C++ with dense $N \times N$ adjacency and edge tensors and no external runtime dependencies. Predictions for the two allele nodes of a variant were averaged, and a variant covered by several windows received the running mean of its predictions. Only variants in the central zone of a window were scored, namely those no further from the trigger than half the window’s largest trigger distance; this is the same zone the network was trained to score, and nodes outside it are read as context but never scored. The attention equations, the edge-context definition, the requirement for the exact erf-based GELU and the central-zone rule are given in Supplementary Method 4; the hyperparameters are in Supplementary Table 3.

Training data. Training windows were produced from LongPhase 2 co-phasing runs on the same HG002 ONT R10.4.1 alignments that the evaluation uses, at the six coverages 10, 12, 14, 16, 18 and 20× and at all ten down-sampling replicates of each, that is from 60 phasing runs. Each run contributed its SNV/indel, SV and 5mC VCFs together with the per-chromosome DOT export, and windows were opened by the same rules as at inference (`-pe_threshold 0.8, -window 20`); the genomic-context features were computed from the same GRCh38 no-alt analysis set. Every window lies within a single chromosome, so the resulting 722,116 windows could be split by chromosome without any window straddling a boundary: chr1–chr14 and chr18–chr20 for training (602,788 windows, 18,356,880 labelled allele nodes in the scored zone), chr15 and chr16 for validation (67,880 windows, 1,509,576 nodes) and chr17, chr21 and chr22 for testing (51,448 windows, 1,367,580 nodes). The training script verifies that the three chromosome sets are disjoint and aborts otherwise. All windows were drawn from the four-class co-phasing configuration, so SV and 5mC records enter the graphs the network was trained on, but only as context: the truth set used for labelling contains small variants alone, so no SV or 5mC node carried a label and among scored nodes only the SNV and indel indicators vary. At inference the module makes no such distinction: a variant’s class enters only through the four indicator features, and SV and 5mC records are scored and unphased on the same footing as SNVs and indels. Corrections to SV and 5mC records are therefore unsupervised: they rest on transfer from the small variants that share their window, and their extent is reported in the Results (Fig. 4c,d).

Labels. Labels were derived from the v5.0q benchmark restricted to chr1–chr22 small variants, the same truth set used for evaluation, and describe the phase of a variant relative to its neighbours rather than in absolute terms. Within each window the variants were grouped by phase set, the first variant of a group present in the benchmark fixed that group’s orientation,

and later variants were labelled by whether their allele-to-haplotype orientation agreed with that anchor. Maximal runs of disagreement were then resolved so that a run is charged to the event that created it, not to each of its members: a single disagreeing variant was labelled *unphase* if its PE was at least 0.9 and *flip* otherwise, and in a run of two or more the first variant, the switch point, was labelled *break* and the remaining members were reset to *correct*. Variants absent from the benchmark, and all SV and 5mC records, were left unlabelled: they were masked from the loss but remained in the graph as context. The released model was trained in binary mode, in which *flip*, *unphase* and *break* are merged into a single *error* class, matching the single action the module is allowed to take. Supplementary Method 4 gives the labelling rules in full.

Class balance and loss. Among the labelled central nodes of the training split, 17,812,710 (97.04%) were *correct* and 544,170 (2.96%) *error*, and 21.5% of windows contained at least one error node. The imbalance was handled by subsampling, not by re-weighting the full distribution: each minibatch kept all of its error nodes and drew correct nodes uniformly at random to at most four times that number (`-ratio_corr 4`), and because the draw was independent at every step the network saw the whole set of correct nodes over training. The cross-entropy on that subset was focal-weighted⁵⁴ by $(1 - p_t)^\gamma$ with $\gamma = 1.5$ and used label smoothing of 0.05. Supplementary Method 4 explains why the error prevalence here is far above the genome-wide switch-error rate.

Training procedure. Each window graph was made undirected before training by adding, for every DOT edge, a reverse edge carrying identical features; the DOT export is strictly one-directional (source position below target position), so without this a node could receive messages only from upstream, and read evidence is in any case symmetric. Two label-preserving augmentations, a haplotype swap and a coverage down-scaling, were applied to training windows only (Supplementary Method 4). Optimization used AdamW⁵⁵ with a peak learning rate of 3×10^{-4} and no warm-up, cosine annealing to 10^{-5} over a 100-epoch horizon, gradient-norm clipping at 1.0, and minibatches of 64 windows batched as disjoint graphs; the remaining optimizer settings are the PyTorch defaults and are listed with the rest of the training configuration in Supplementary Table 4. Dropout of 0.4 was applied to the output of the local branch, to the output of the global branch and inside each feed-forward block; dropout is inactive at inference and therefore does not appear in the exported graph. The run was seeded once (`-seed 42`, seeding both the Python and the PyTorch generators) and stopped early after 25 epochs without improvement of the validation criterion, at epoch 69 of the 100-epoch budget. Training used PyTorch 2.11.0 with PyTorch Geometric 2.7.0 under Python 3.12, CUDA 13.0 and cuDNN 9.19, on one NVIDIA GeForce RTX 5090. All reported *gnn* results come from this single training run; sensitivity to reseeding was not assessed.

Model selection. Checkpoints were selected on the validation chromosomes by a net-gain criterion instead of validation loss or F1: at each epoch the predicted error probability was thresholded over 81 values spanning [0.01, 0.99] and the score $S = R(2 - 1/P)$ was maximized subject to a recall of at least 0.005, where P and R are the precision and recall of the error class. S is the net fraction of true errors removed when a detection is worth one error and a false detection costs one, and is positive only for $P > 0.5$. The best checkpoint was epoch 44 ($S = 0.0235$, $P = 0.626$, $R = 0.058$, threshold 0.868). On the held-out chromosomes chr17, chr21 and chr22 it attained an accuracy of 0.941 with error-class precision, recall and F1 of 0.258, 0.499

and 0.340 under the argmax decision, over 41,378 labelled error nodes. The threshold applied by `gnn`, 0.30, is far more permissive, because the module’s only action is to withhold phase: a false detection costs one phased variant rather than one switch error. Supplementary Method 4 gives the full selection procedure. [State how 0.30 was chosen: a sweep, on which data and against which objective, or an a-priori choice. Add the phased-fraction/switch-error trade-off across thresholds as a supplementary panel.]

Export and numerical agreement. The trained network was re-implemented with dense tensors, exported to ONNX and embedded in the binary as a base64-encoded float32 blob, and the forward pass refuses to run if the blob length disagrees with the expected parameter count. Numerical agreement was verified at each link of that chain; the largest observed difference in predicted error probability was 3×10^{-7} , between the compiled C++ and ONNX Runtime on six synthetic graphs of 3–120 nodes. Supplementary Method 4 describes the export chain and the acceptance bounds.

Hold-out design. The hold-out is at the level of the chromosome. No window from chr17, chr21 or chr22 entered any gradient, any batch-normalization statistic or the checkpoint selection, and chr15 and chr16 were used only for selection. Nothing else was held out. The training windows come from the same 60 down-sampling replicates at 10–20× that Figs. 2–5 evaluate, the same v5.0q benchmark supplied both the training labels and the evaluation truth, and the figures are computed genome-wide over chr1–chr22, so 17 of the 22 evaluated chromosomes contributed training windows. No training window was drawn above 20×, although results are reported to 60×, and all training and evaluation data come from one individual (HG002) on one platform and chemistry. The consequences for generalization are discussed in the Discussion.

Correction. A variant whose mean predicted error probability was ≥ 0.30 (`-break-threshold`) was unphased: its genotype was rewritten in unordered, ascending form (0|1 and 1|0 both become 0/1) and its PS tag was removed, while the pair of alleles was preserved and no genotype was ever flipped. Bridge vertices may optionally be protected (`-respect-bridge`; off by default). After unphasing, the remaining phased variants of each phase set were checked for connectivity through the surviving DOT edges; when a phase set became disconnected, the largest component kept the original PS and every other component received a new PS equal to the 1-based position of its first variant (`-no-split-blocks` disables this). The corrected SNV/indel, SV and 5mC VCFs are written separately, as `<prefix>.vcf`, `<prefix>_SV.vcf` and `<prefix>_mod.vcf`.

Haplotagging

Reads were assigned to haplotypes by `haplotag`. For each primary alignment with mapping quality ≥ 1 (`-qualityThreshold`), and for supplementary alignments under `-tagSupplementary`, the module counted the phased SNV, indel, SV and 5mC alleles the read carries on each haplotype, every informative observation contributing one vote regardless of class, and tagged the read with `HP:i:1` or `HP:i:2`, the phase set PS and a Phred-scaled phasing quality PQ when the majority haplotype accounted for at least 60% of the informative alleles (`-percentageThreshold`). Two conditions left a read untagged: no clear majority, and informative variants belonging to more than one phase set, for which a majority has no meaning. SV alleles were re-derived from

the CIGAR string with the same window and length tolerance as in **phase**, and 5mC evidence was matched by read name from the **MR** list of the phased 5mC VCF. The evidence model, which weights the four variant classes equally and therefore differs from the read-based correction inside **phase**, and the definition of **PQ** are given in Supplementary Method 5.

Benchmark data

HG002 (NA24385) Oxford Nanopore R10.4.1 reads were those of the Oxford Nanopore Technologies open-data release of the GIAB Ashkenazi trio sequenced with Ligation Sequencing Kit V14 (`giab_lsk114_2022.12`): two PromethION R10.4.1 flow cells (FLO-PRO114M; PAG65784 and PAG68757) run with SQK-LSK114 at 400 bases per second and 4 kHz sampling in November 2022, basecalled in simplex mode by Oxford Nanopore with Dorado using the `dna_r10.4.1_e8.2_400bps_sup@v4.0.0` model and the `5mCG_5hmCG@v2` modified-base model, with a read N50 of 29.2 and 29.4 kb for the two flow cells (MinKNOW run reports). The reads were aligned to the GRCh38 no-alt analysis set without decoys (`GCA_000001405.15_GRCh38_no_alt_analysis_set`) with minimap2 v2.30 using `-ax map-ont -y`. The full data set (mean autosomal depth 70.4×, from the mosdepth depth distribution released with Oxford Nanopore’s alignment of the same reads to the same reference) was down-sampled with `samtools view -s` (`samtools v1.23.1`)⁵⁶ to 10, 12, 14, 16, 18, 20, 30, 40, 50 and 60×; for 10–20×, ten independent random seeds were drawn and all metrics are reported as the mean over the replicates, whereas 30–60× used a single replicate ([give the ten seed values, or state that they were 1–10, so the down-sampling is reproducible]). Variants were called separately on every down-sampled alignment, for each coverage and each replicate, so that no call set was carried over from another coverage: SNVs and indels with Clair3 v2.0.1 using the model `r1041_e82_400bps_sup_v500`[Check this model. The reads are a 4 kHz `sup@v4.0.0` basecall, but `r1041_e82_400bps_sup_v500` is trained on 5 kHz Dorado v5.0.0 SUP data. The matched model is `r1041_e82_400bps_sup_v400` (Rerio), or `sup_v410`, the 4 kHz model bundled with Clair3. Correct it if another model was run; if `sup_v500` was run, expect a reviewer to ask why.], SVs with Sniffles2 v2.8.0 with `-output-rnames`, and allele-specific 5mC sites with `longphase modcall`. Phasing accuracy was assessed against two truth sets: the GIAB v4.2.1 small-variant benchmark for HG002³⁵ (`HG002_GRCh38_1_22_v4.2.1_benchmark.vcf.gz` with `HG002_GRCh38_1_22_v4.2.1_benchmark_noinconsistent.bed`, from the GIAB FTP release directory `ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/`), and the small-variant benchmark that GIAB derives by aligning the complete diploid T2T-HG002 assembly (Q100 project, T2T-HG002 v1.1)^{32,36,57} to GRCh38, distributed as a draft of the GIAB v5 release³³ and referred to throughout this paper as v5.0q. The T2T-HG002-derived benchmark adds 701 Mb of autosomal sequence that v4.2.1 does not assess, most of it satellite and segmentally duplicated³². Identical phased VCFs were scored against both truth sets, each restricted to the regions of its own benchmark BED and to chr1–chr22, so chrX, chrY and chrM enter no metric.

Comparison with other methods

WhatsHap v2.8⁷ and HapCUT2 v1.3.4⁸ were run on the same alignments and the same Clair3 calls ([give the complete command line for each, including WhatsHap phase options such as `-ignore-read-groups` and any `-indels/-distrust-genotypes` setting, and the HapCUT2 two-step invocation `extractHAIRS -ont 1` followed by `HAPCUT2`, stating whether indels were phased by either tool and whether HapCUT2’s default likelihood-based pruning was left on; “default parameters” is not reproducible for either tool.]). LongPhase 1.0¹⁰ (release tag v1.0, commit fe8c1fdff11ba2a082d363a7ffcb5c923cacc316) and LongPhase 2 were run with `-ont`; the GNN module was applied to the LongPhase 2 output with default thresholds ($PE \geq 0.8$, break threshold 0.30, no bridge protection, block splitting enabled). For co-phasing experiments, the SNV, SNV+indel, SNV+SV, SNV+5mC and SNV+indel+SV+5mC configurations were phased separately, and the GNN was applied to each. Cross-tool comparisons used the SNV-only configuration of LongPhase 2 and were scored on SNVs. WhatsHap and HapCUT2 can phase indels but not SVs or 5mC sites, so indel, SV and 5mC co-phasing were evaluated as LongPhase 2 with and without GNN correction. [State that all tools ran on the same machine, and whether WhatsHap, HapCUT2 and LongPhase 1.0 were re-run on every down-sampling replicate at 10–20× (the LongPhase 2 tables hold every replicate).]

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed with `longphase compare`, a C++ re-implementation of the `whatshap compare` algorithm⁷ that is parallelized over chromosomes with OpenMP. For each chromosome, heterozygous variants common to the truth and query VCFs were intersected by position and alleles and partitioned into blocks that are phased in both files; singleton blocks were ignored. Within each block the query haplotype string was compared with the truth string in switch encoding (the sequence of relative phase changes between consecutive variants), giving the number of *switch errors*; the *switch error rate* is the number of switch errors divided by the number of assessed variant pairs. The *Hamming distance* is the block-wise minimum over the two possible haplotype orientations of the number of misassigned variants, summed over blocks and expressed as a percentage of compared variants. Switch errors were further decomposed into isolated switches and flips (two consecutive switches). The *phased fraction* is the number of truth heterozygous variants phased by the query divided by the total number of truth heterozygous variants, reported separately for SNVs, indels, SVs and 5mC sites; the *block N50* is the span-weighted median block length of the query, pooled over all chromosomes and independent of the truth set. Switch-error rates and Hamming distances for cross-tool comparisons were computed on SNVs only, so that the co-phasing of additional variant classes did not change the evaluated set. Switch-error positions were exported as BED and used for the analysis of unphased variants in Fig. 6. Supplementary Method 6 defines each metric precisely. [Define the depth and informativeness bins used in Fig. 6b (“low”, “normal”, “high” relative to the genome-wide mean, and the informativeness cut-points) and the “low genotype quality” threshold used in Fig. 6c.]

Runtime and memory

Wall-clock time and peak resident memory were measured with `/usr/bin/time -v` on [CPU model and clock, physical core and socket count, RAM, storage type, operating system and kernel, compiler and optimization flags], using $[t]$ threads for every tool. [State whether the machine was otherwise idle, how many repeats were timed and whether the reported value is the mean, median or minimum, and report peak resident memory alongside every runtime, since the Results section currently quotes runtimes only and Fig. 2e has no memory counterpart.] GNN correction time was measured separately and added to the LongPhase 2 total. Reported LongPhase 2 times include reading of the alignments, graph construction, phasing and writing of the VCF, and exclude variant calling, alignment and down-sampling.

Statistics and reproducibility

No statistical tests were used to compare methods, and no sample-size calculation, randomization or blinding was applicable: all metrics are deterministic functions of the input data and the only stochastic element is the choice of reads during down-sampling. The replicate design was as follows. At 10, 12, 14, 16, 18 and 20 $\times$, ten independent down-sampling replicates were generated from the same full-coverage alignment with ten distinct random seeds, and every reported value at those coverages is the mean over those replicates. At 10 $\times$ the seed-10 replicate is not available, so every value quoted at 10 $\times$, and every 10 $\times$ point in Figs. 2–5, is a mean over $n = 9$ replicates; all other coverages in that range use $n = 10$. At 30, 40, 50 and 60 $\times$, a single replicate was generated and analysed, so no dispersion can be reported at those coverages and differences between methods there are not accompanied by a measure of replicate variability. [Required before submission: the LongPhase 2 values quoted at 10 $\times$ in the Results now carry the replicate s.d. from the `compare` tables; add the same for the WhatsHap, HapCUT2 and LongPhase 1.0 values once their tables are available, and draw the dispersion in Figs. 2–5, as either s.d. or the full range over the replicates, saying in each legend which is drawn and giving n . (Further replicates at 30–60 $\times$ would let the high-coverage ranking carry a dispersion estimate; optional, since the legends and this paragraph already state that those points are single replicates.)] The output of `phase` does not depend on the thread count by construction: the program parallelizes over chromosomes, each chromosome is processed from start to finish by one thread, and the HTSlib thread pool serves only to decompress the alignments, which are consumed in file order (`PhasingProcess.cpp:96`; `ParsingBam.cpp:1012`). `compare` likewise evaluates every intersection block independently and sums integer counts. The GNN module performs no sampling and uses fixed weights, so its output is deterministic up to floating-point summation order: the error probability of a variant covered by several windows is averaged in the order in which threads complete, which can change the last bits of the mean (of order 10^{-7}) and hence the decision only for a variant whose mean lies within that distance of the 0.30 threshold.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The GRCh38 no-alt analysis set used as the reference is distributed by NCBI as GCA_000001405.15_GRCh38_no_alt_analysis_set.fna.gz (GenBank assembly GCA_000001405.15), at https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/. The GIAB HG002 v4.2.1 small-variant benchmark (HG002_GRCh38_1_22_v4.2.1_benchmark.vcf.gz and HG002_GRCh38_1_22_v4.2.1_benchmark_noinconsistent.bed) is at https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/. The v5.0q benchmark is the GRCh38 small-variant call set of the GIAB draft benchmark release NIST_HG002_DraftBenchmark_defrabbV0.020-20250117 (GRCh38_HG2-T2TQ100-V1.1_smvar.vcf.gz, with the accompanying GRCh38_HG2-T2TQ100-V1.1_smvar.benchmark.bed), at https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/data/AshkenazimTrio/analysis/NIST_HG002_DraftBenchmark_defrabbV0.020-20250117/; the file scored here, HG002_GRCh38_v5.0q_smvar.chr1_22.vcf, is the chr1–chr22 subset of that VCF and was scored within the accompanying BED. The HG002 nanopore reads are from the Oxford Nanopore open-data release giab_lsk114_2022.12 (s3://ont-open-data/giab_lsk114_2022.12/; raw signal under flowcells/hg002/, Dorado sup@v4.0.0 basecalls with modified-base tags and their alignment under analysis/wf-human-var-output/hg002_sup_v4/), distributed under the Creative Commons Attribution-NonCommercial 4.0 International licence (CC BY-NC 4.0). [Still required. (i) State the command used to derive the chr1–chr22 subset of the v5.0q VCF. (ii) Source data for Figs. 2–6 and for every Supplementary Figure with plotted values. Nature Methods requires that restrictions, if any, are stated; if there are none, say so explicitly.]

Code availability

LongPhase 2, including the modcall, phase, gnn, haplotag and compare sub-commands, is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>. A Dockerfile, pre-compiled Linux x86-64 binaries and a demonstration data set (chr1:121.5–122.0 Mb) are provided in the repository, together with a script that runs modcall, SNV-only and four-class phase, gnn correction of both and compare against a truth VCF in under two minutes on four threads. [Give the exact release tag and the full 40-character commit hash of the version used for every result in this paper. The latest tagged release at the time of writing is v2.0.2 (main.cpp:9; README.md:34), but v2.0.2 predates the gnn and compare sub-commands, which exist only on the unmerged JH branch at commit cc17fb1, so no released version can currently reproduce any gnn or compare result in this paper. Merge and tag before submission.] The complete training chain is provided as a separate archive, comprising the code for window and feature extraction, training and model selection, ONNX export and weight em-

bedding, together with the driver scripts, the released checkpoint in both PyTorch and ONNX form, the training log and a pinned list of package versions. [These files exist but are not yet in the public repository or in any archive. Deposit them in the repository under an OSI-approved licence and cite them here; Nature Methods will not accept a machine-learning method whose training code is unavailable. Note that the scripts as written carry absolute paths of the development machine (/ssd/longphase_process/..., /disk/software/...); parameterize them before deposition.] [No script currently produces Figs. 2–6: they are raster exports of slides. Write the plotting script that reads the longphase compare tables and deposit it with the tables. The schematic Supplementary Figs. 1–8 are generated by figures/supp/make_supp_figs.py in the manuscript repository and can be deposited as is.]

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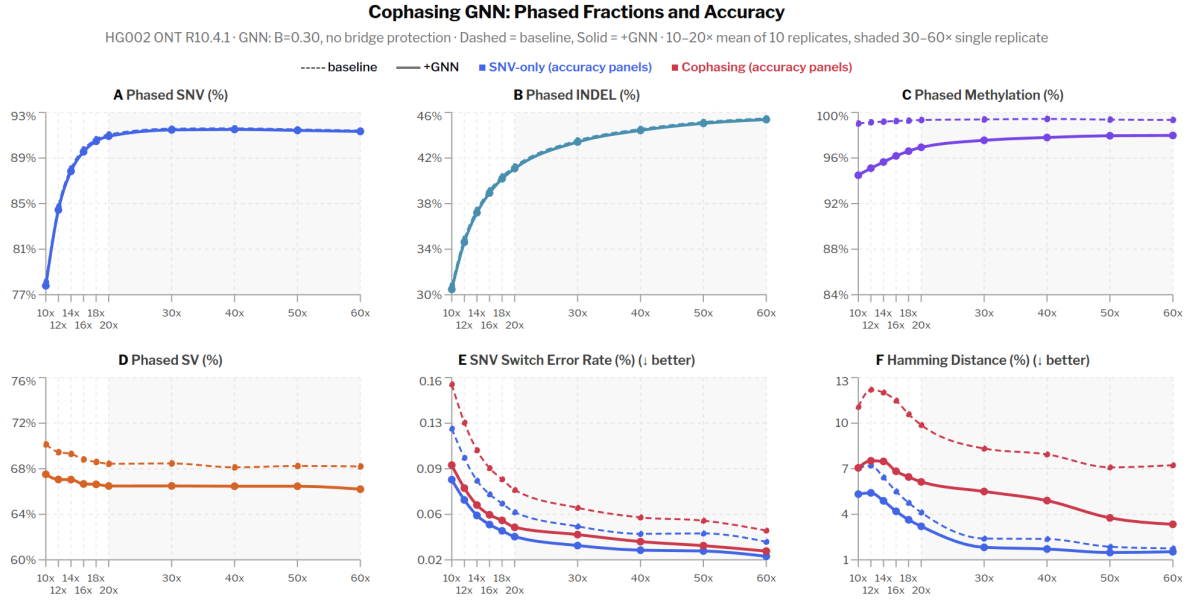


Figure 4: **Co-phasing of SNVs, indels, SVs and allele-specific methylation.** HG002 nanopore R10.4.1 data, 10–60×; replicates as in Fig. 2; dashed lines, LongPhase 2; solid lines, LongPhase 2 with neural-network correction. **a–d**, Fraction of phased SNVs (**a**), indels (**b**), allele-specific 5mC sites (**c**) and SVs (**d**) in the four-class co-phasing configuration. **e,f**, SNV switch error rate (**e**) and Hamming distance (**f**) for SNV-only phasing (blue) and four-class co-phasing (red), each with (solid) and without (dashed) correction. Denominators: GIAB v5.0q heterozygous SNVs and indels; Sniffles2 heterozygous SVs; modcall heterozygous 5mC sites.

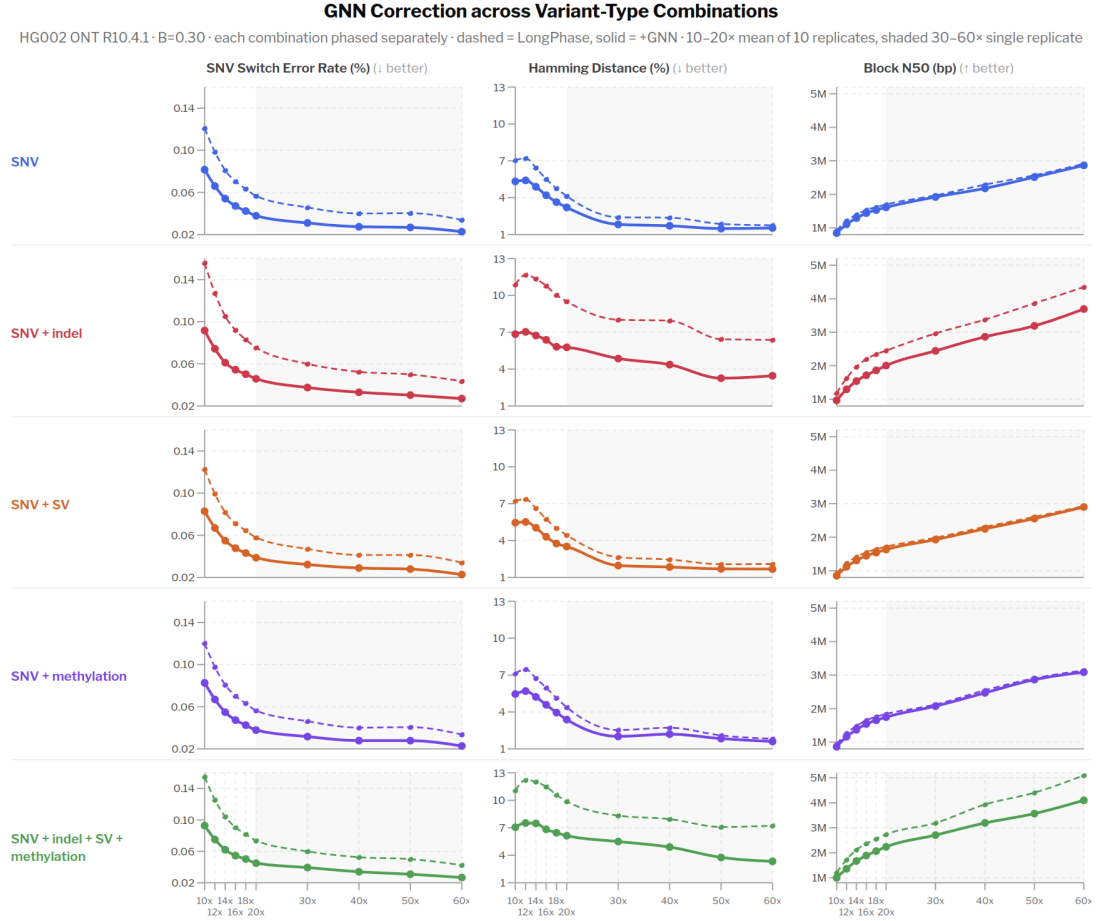
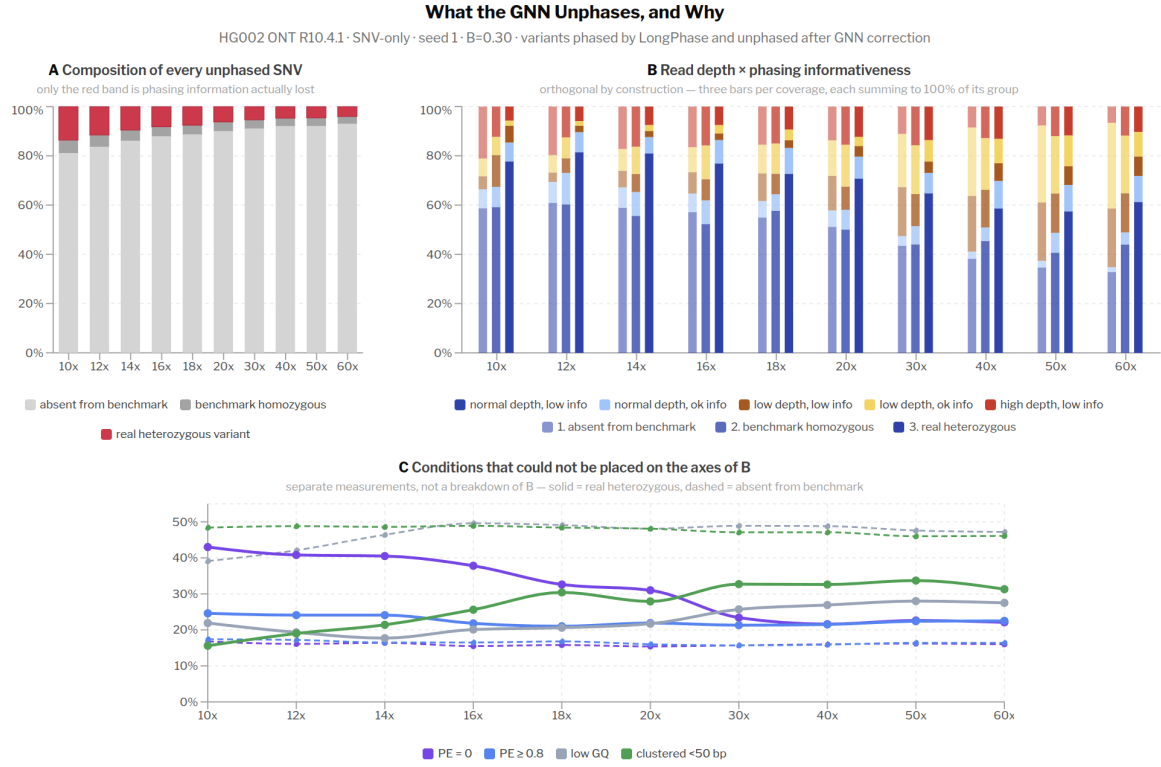


Figure 5: **Effect of neural-network correction across variant-class combinations.** Each row is a separate phasing run of HG002 nanopore R10.4.1 data with the indicated variant classes; dashed, LongPhase 2; solid, LongPhase 2 with correction. Columns: SNV switch error rate, Hamming distance and phase-block N50, scored against the v5.0q benchmark on SNVs. Replicates as in Fig. 2.



The GNN unphases roughly 60,000 SNVs at every coverage, but 86–96% of them are calls the benchmark does not support as heterozygous, so almost none of the apparent loss is real phasing information (A); at 60× only 4.0% (2,411 variants) are genuine heterozygous sites. Placing each variant on two orthogonal axes (B) shows that true heterozygotes are removed overwhelmingly at normal read depth with weak phasing informativeness (61–81%), whereas benchmark-absent calls shift towards low depth as coverage rises (12.5% at 10× to 58.6% at 60×), the signature of hard-to-map regions. No variant is unphased for high depth alone: the high-depth / normal-informativeness cell is empty at every coverage. Panel C reports conditions that overlap too heavily to sit on those axes, as plain prevalences within each group. Among the true heterozygotes the GNN removes, 43.0% at 10× have a PE of exactly zero — against 16.8% of benchmark-absent calls — so no fixed entropy threshold could have found them; that share falls to 22.1% by 60× as phasing evidence improves. Clustering and low genotype quality run the other way, staying near 46–49% among benchmark-absent calls at every coverage while remaining well below that among true heterozygotes, which is what marks them as caller artefacts rather than phasing failures.

Figure 6: Composition of the variants unphased by the neural network. SNV-only phasing of HG002 nanopore R10.4.1 data (one down-sampling replicate, seed 1; break threshold 0.30); every SNV phased by LongPhase 2 and unphased after correction (~60,000 per coverage) is classified against the v5.0q benchmark. **a**, Fraction of unphased SNVs that are absent from the benchmark, homozygous in the benchmark or genuine heterozygous sites; only the last category represents lost phasing information. **b**, Within each of the three categories, distribution over read depth (low, normal, high relative to the genome-wide mean) and phasing informativeness (number of informative linking reads); the three bars at each coverage each sum to 100% of their category. **c**, Prevalence of conditions that overlap the axes of **b**: phasing entropy of exactly 0, phasing entropy ≥ 0.8 , low genotype quality and clustering within 50 bp of another variant; solid, genuine heterozygous sites; dashed, benchmark-absent calls.